

Partial Differential Equations PhD Exam, May 2010

1. Suppose $u \in C_1^2(\mathbb{R}^n \times [0, T]) \cap C(\mathbb{R}^n \times [0, T])$ solves

$$\begin{cases} u_t - \sum_{i,j=1}^n a_{ij} \partial_{ij} u = 0, & \text{in } \mathbb{R}^n \times (0, T], \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

where $(a_{ij})_{n \times n}$ is a positive definite, symmetric constant matrix, g is a given smooth function. Suppose in addition that

$$u(x, t) \leq A e^{a|x|^2}, \quad x \in \mathbb{R}^n, \quad t \in [0, T]$$

for some $a, A > 0$. Prove

$$\sup_{\mathbb{R}^n \times [0, T]} u = \sup_{\mathbb{R}^n} g.$$

2. Let u solve

$$\begin{cases} u_{tt} - u_{xx} = 0, & \mathbb{R}_+ \times (0, \infty), \\ u = g, \quad u_t = h, & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ u = 0, & \text{on } \{x = 0\} \times (0, \infty). \end{cases}$$

where g, h are smooth, given functions of x such that $g(0) = h(0) = 0$. Write the explicit expression of u .

3. Let B_1 be the unit ball in $\mathbb{R}^n$ ($n \geq 3$), $p \in (1, \frac{2n}{n-2})$ and $\alpha \in (0, 1)$, show that there exists a C depending only on n, p, α such that

$$\left(\int_{B_1} |u|^p dx \right)^{1/p} \leq C(n, p, \alpha) \left(\int_{B_1} |\nabla u|^2 \right)^{1/2},$$

for all $u \in H^1(B_1)$ that satisfy the following property:

$$|\{x \in B_1; u(x) = 0\}| \geq \alpha |B_1|.$$

4. Suppose $u \in C^2$ solves

$$u_{tt} - \Delta u = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty).$$

Fix $x_0 \in \mathbb{R}^n$, $t_0 > 0$ and consider the cone

$$C = \{(x, t); 0 \leq t \leq t_0, |x - x_0| \leq t_0 - t\}.$$

Prove that if $u \equiv u_t \equiv 0$ on $B(x_0, t_0) \times \{t = 0\}$, then $u \equiv 0$ in C .

5. Let $(a_{ij}(x))_{n \times n}$ be smooth, symmetric on B_1 and there exist $0 < \lambda < \Lambda$ such that

$$\lambda |\xi|^2 \leq \sum_{i,j=1}^n a_{ij}(x) \xi_i \xi_j \leq \Lambda |\xi|^2, \quad \forall x \in B_1, \quad \forall \xi \in \mathbb{R}^n.$$

Let b_i ($i = 1, 2, \dots, n$), c and f be smooth functions on B_1 . Show that there exists $\delta > 0$ such that if

$$\|b_i\|_{L^\infty(B_1)} + \|c\|_{L^\infty(B_1)} \leq \delta, \quad i = 1, \dots, n,$$

then there exists a smooth function u that solves the Dirichlet boundary problem uniquely:

$$\begin{cases} -\sum_{i,j=1}^n \partial_i(a_{ij}(x)\partial_j u) + \sum_{i=1}^n b_i(x)\partial_i u(x) + c(x)u(x) = f, & B_1, \\ u = 0, & \text{on } \partial B_1. \end{cases}$$

6. Let $\{u_m\}$ be a sequence of smooth functions satisfying $u_m(x) \leq C$ and $|\Delta u_m| \leq C$ for some constant C uniformly over B_1 , the unit ball. Prove that over $B_{1/2}$, along a subsequence, either u_m converges to $-\infty$ uniformly, or u_m converges to a function u in C^2 norm.

7. **Warning: Suspected error.** The displayed variational form places the lower-order terms inside the scope of the outer sum over i, j , while also using an inner sum over i . This potentially unintended summation scope has been retained because the intended correction is not uniquely determined by the immediate context.

Let $\{a_{ij}\}_{n \times n}$ ($n \geq 2$) be a positive definite, symmetric, constant matrix, $b_1, \dots, b_n, c$ be smooth functions on B_1 , the unit ball in $\mathbb{R}^n$. Show that the minimum of the following variational form

$$I(w) = \int_{B_1} \sum_{i,j=1}^n \left(a_{ij} \partial_i w \partial_j w + \sum_i b_i \partial_i w w + c w^2 \right) dx, \quad w \in H_0^1(B_1)$$

under the constraint: $\int_{B_1} |w|^2 = 1$ can be reached by a smooth function.